Pathogen genome analysis with PathogenWatch

Version: April 2025

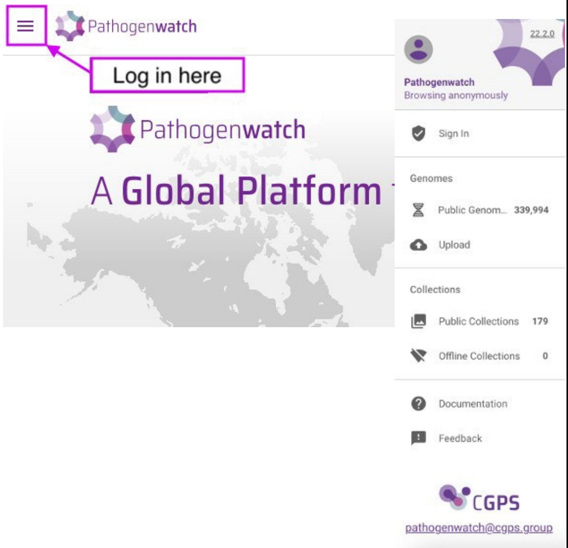
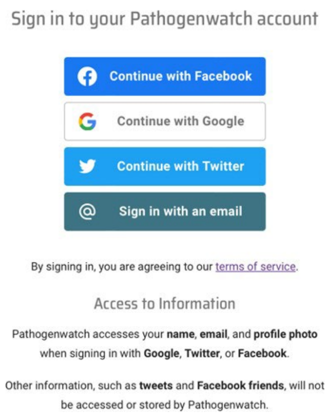
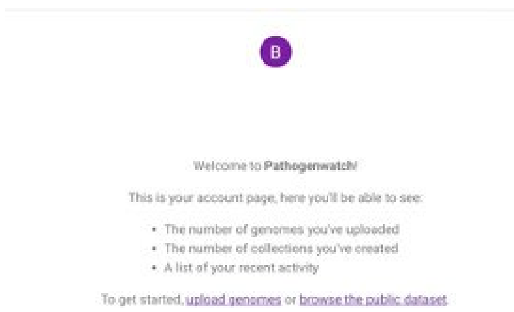
Utrecht University

Introduction

Pathogenwatch (https://pathogen.watch/) is a web-based tool for genomic surveillance that enables users to upload and analyse their own genome data. Pathogenwatch performs a series of species-specific analytics including MLST, identification of resistance and virulence loci, replicon typing and genome clustering. MLST Sequence typing is available for over 100 species using schemes from PubMLST, Pasteur, and Enterobase. Users can create interactive visualisations of phylogenetic trees make with genome collections of publicly available genomes and including their own uploaded genomes data.

Manual to upload genomic data to Pathogenwatch

Create an PathogenWatch account

Open PathogenWatch on your browser https://pathogen.watch/ Click on “Sign In” by clicking on the icon in the upper left corner of the page. Note: it is possible to set your metadata private, see https://cgps.gitbook.io/pathogenwatch/how-to-use-pathogenwatch/private-metadata	
Create your own login credentials Check your SPAM folder if you choose to sign in by email and have difficulties receiving the login link.	
View My Account page Now that you are logged in, you will see your Account page where you can manage your genomes, collections and account settings.	

Navigating PathogenWatch

<table>
<tr>
<td>

Navigate the site

- Click the menu icon

The main menu bar can be accessed by clicking on the menu con on the upper left corner of the page.

</td>
<td>

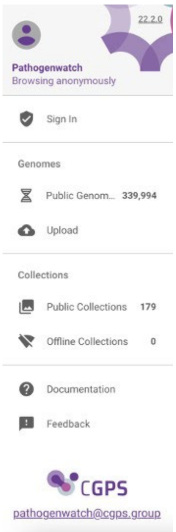
</td>
</tr>
<tr>
<td>

The menu allows you to navigate the site. There are four key pages:

- Genomes
- Collections
- Upload
- Documentation

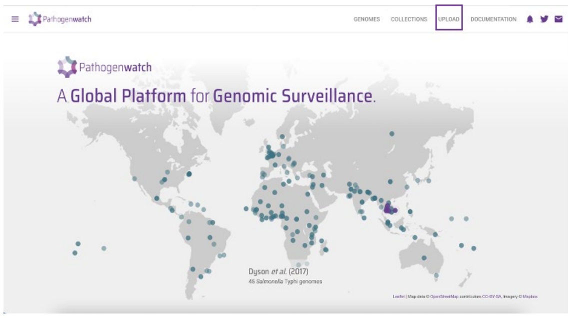
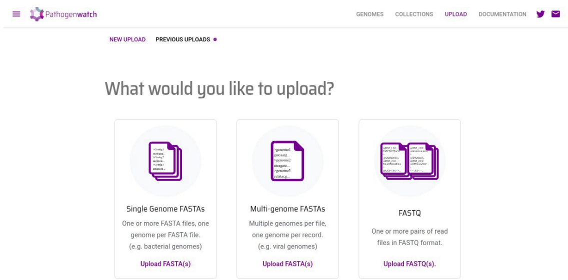
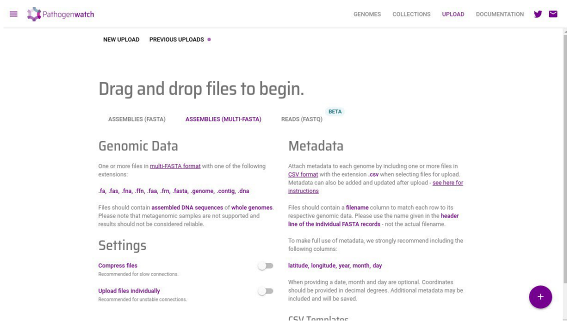
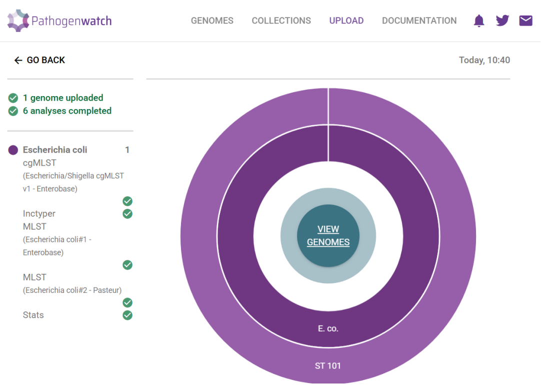
More information about navigating the site: https://cgps.gitbook.io/pathogenwatch

</td>
<td>



</td>
</tr>
</table>

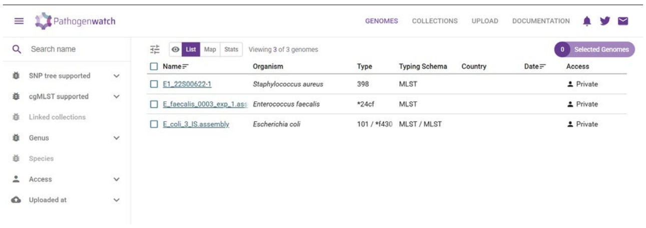
Importing data

Navigate to the “upload” page: On the home screen, click on “Upload” in the upper right corner.	
Select the file type to upload Click the “Upload” button Click on “Single Genome FASTAs”	
Drag and drop files to begin Click the “+” button to select files to upload	
Processing screen and viewing results Depending on the species of the genome, Pathogenwatch will give the following results: - Species - AMR (not for every species) - Core summary (not every species) - cgMLST - Inctyper (not for every species) - MLST If you click the "View Genomes" button, you will be directed to the results page of all your uploaded genomes.	

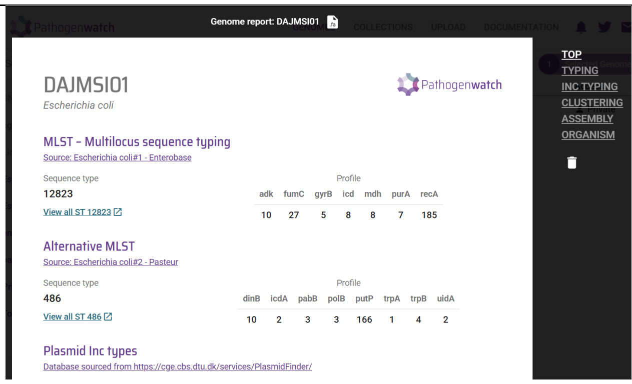
Viewing genomes

View your uploaded genomes

The uploaded assemblies are presented in the "view Genomes" page



You can click the names of the uploaded genomes.
This will show you the "genome report", a detailed report of PathogenWatch analysis of the assembled genome.



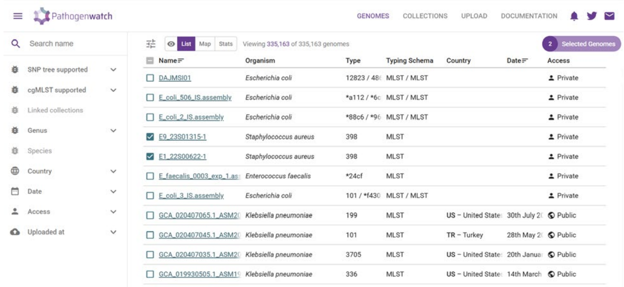
Creating a tree with your genome and publicly available genomes

Note: only a selection of species can be used to build trees. Unfortunately, it is not possible to build a tree with *E. coli*.

In this example, we will use *Staphylococcus aureus*

Go to the page with the list of your uploaded genome(s) and select the genome that you want to include in the tree

Select only genomes of similar species/MLST STs.
For different species/STs; build different trees



<table>
<tr>
<td>

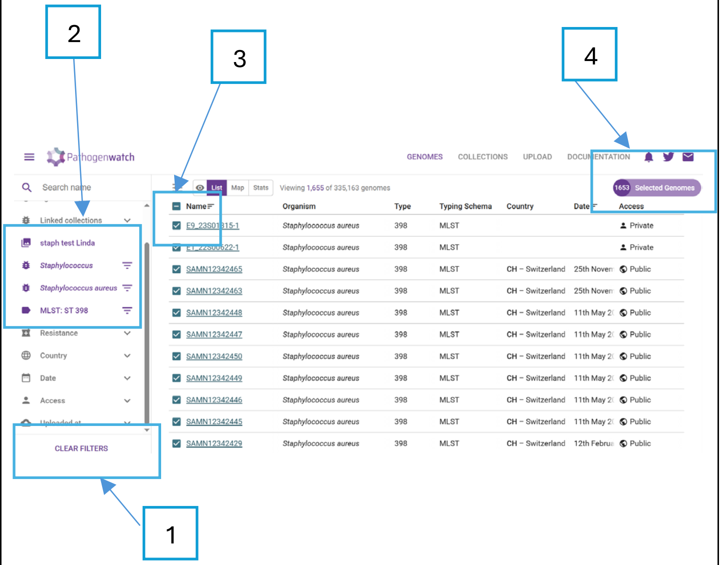
1. Click on “CLEAR FILTERS” at the bottom left corner.

2. Filter the publicly available genomes of Pathogenwatch from the left sidebar. Filter the genomes with the analysis results of your genome. Next is an example:
    - Genus – Staphylococcus
    - Species – Staphylococcus aureus
    - MLST – ST398

3. Click on the tick that selects all genomes and then click on “Selected Genomes”.

4. You can see that the number of genomes has now increased.

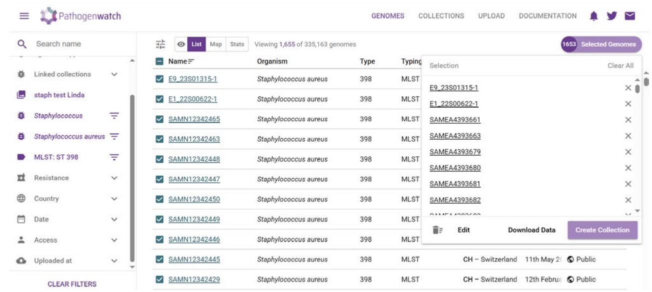
</td>
<td>



</td>
</tr>
<tr>
<td>

Create a collection with your genome(s) and the publicly available selected genomes by clicking on “Selected genomes” in the upper right corner and select “Create collection”

</td>
<td>

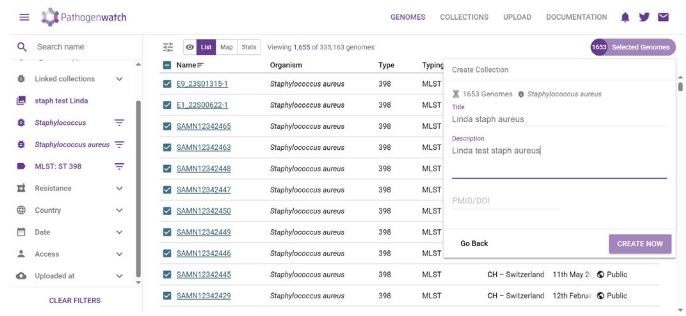


</td>
</tr>
<tr>
<td>

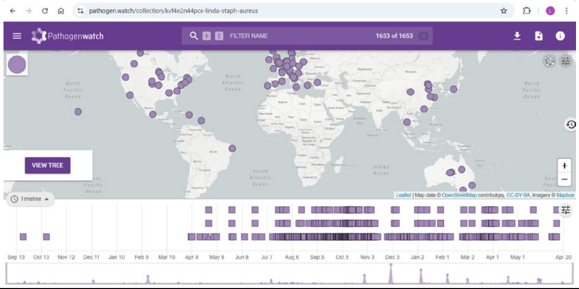
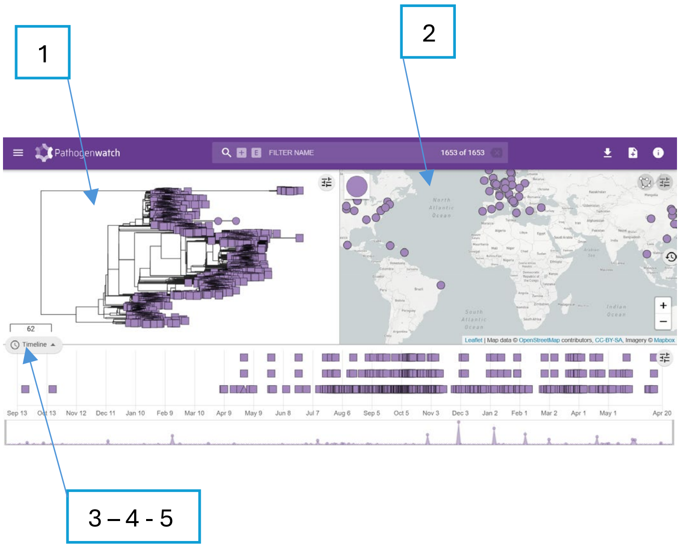
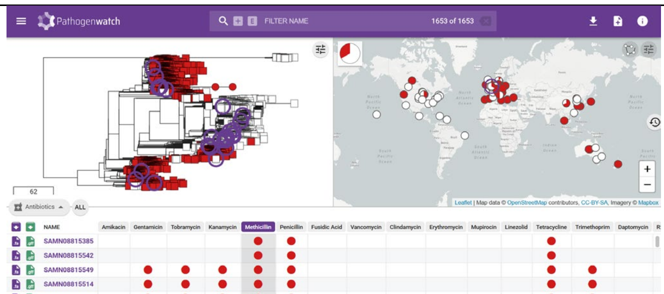
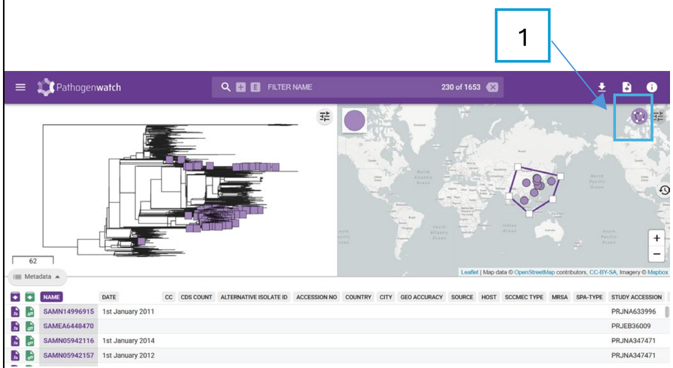
Name your collection as follows:
- Title
- Description - A short description of the collection, perhaps to explain to other people what it is for.
- PMID/DOI (optional) The PubMed or DOI identifier if it is linked to a publication.

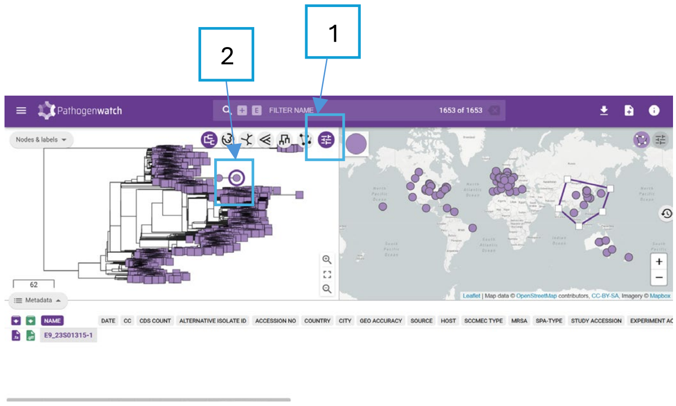
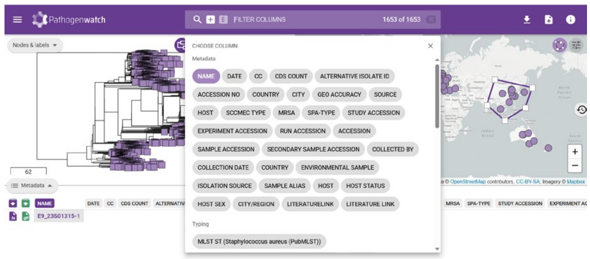
Click "Create Now" and you will be taken to the “Collection View”.

</td>
<td>



</td>
</tr>
</table>

If you click “view tree”, a tree will be calculated using a neighbour-joining method	
Locate the five interactive components: 1. A phylogenetic tree. 2. A global map showing the locations of genomes that have geospatial metadata. 3. A metadata display area that includes the user-supplied data, calculated typing assignments such as MLST, and antimicrobial resistance predictions. 4. A query bar that allows the selection or finding of assemblies by searching the metadata fields. 5. A timeline showing the collection dates of assemblies that have temporal metadata.	
In the “Antibiotics” window, you can select a specific antibiotic. Circles in the map will be colored present/absent (red/white) genotype or resistant/sensitive (red/white) phenotype. The circle in the top left shows the overall ratio for the selected resistance phenotype or genotype	
With the group selection tool (1), you can draw a line around a set of points to select. These are then highlighted in the “Tree View” and “Metadata Table”.	

1. Click on this icon to see all different trees, zoom function etc. 2. You can click on leaves and nodes and they are selected with a purple halo. Selected assemblies are highlighted in the “Map View” also with a purple halo and shown exclusively in the “Metadata Tables”. Tip: If you right click on the tree, you can export the tree, labels etc.	
The “Filter Bar” allows you to select assemblies according to their attributes in the “Metadata Tables”, including by typing assignments or antimicrobial resistance. You can filter on multiple columns by clicking on the '+' symbol.	
Download tree and files 1. In the upper right corner, you will find the download button, where you can download metadata tables, profiles and tree files. 2. To download the fasta and/or gff files, select the files in the metadata window.